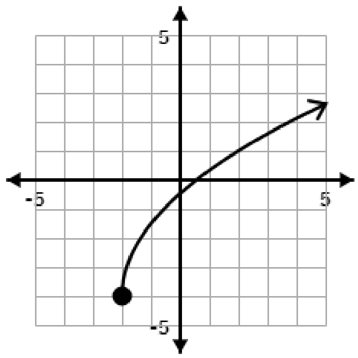
Identify the transformations and write the equation of the graph shown



Given $f(x) = x^2 - 10x + 27$, evaluate $f(5 - 2i)$.

What is notable about this result?

Given $f(x) = x^2 + 12x + 54$, find

a) the vertex of f

b) the roots of f

Draw the graph of the parabola given by

$$(y - 3)^2 = -6(x + 1)$$

and give the focus and directrix

Given the general form conic equation $9x^2 + 5y^2 - 20y - 160 = 0$

a) Identify the conic section

b) Write the equation in standard form

Given the general form conic equation $x^2 - 2y^2 - 6x - 4y + 25 = 0$

a) Identify the conic section

b) Write the equation in standard form

An augmented matrix is given below

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 \\ 3 & 1 & 6 & 0 & 0 & 1 \end{array}\right]$$

Perform the following row operations:

- $R_3 = 3R_1 - R_3$
- $R_2 = \frac{1}{2}R_2$
- $R_3 = 4R_2 + R_3$

(if you finish early, see if you can get the identity matrix on the left side)

The general form equation of a circle can be written $x^2 + y^2 + Dx + Ey + F = 0$.

Write a matrix equation of the form $[A][X] = [B]$ that could be used to find the general form equation of a circle passing through the points (0, 4), (1, –2), and (2, 2).

If you have time, use a calculator to solve and write the general form equation.

Question: Given $f(x) = x^2 - 10x + 27$, evaluate $f(5 - 2i)$. What is notable about this result?

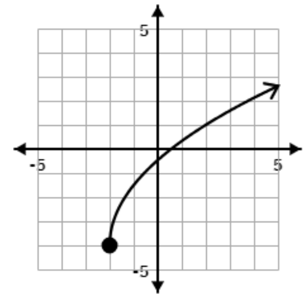
Answer:

$$f(5 - 2i) = (5 - 2i)^2 - 10(5 - 2i) + 27$$
$$= 25 - 20i + 4i^2 - 50 + 20i + 27$$
$$= 2 + 4(-1) = -2$$

This result is notable because a complex domain value output a real range value.

Question: Identify the transformations and write the equation.

Answer:

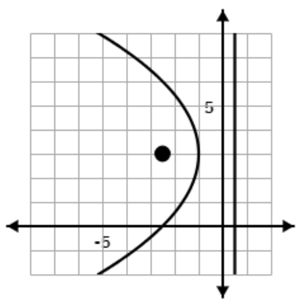


Vertical stretch by a factor of 2.5; shift left 2; shift down 4

$$f(x) = \frac{5}{2}\sqrt{x + 2} - 4$$

Question: Draw the graph of the parabola given by $(y - 3)^2 = -6(x + 1)$ and give the focus and directrix.

Answer:



We notice that $4p = -6$, so $p = -1.5$

Focus: $(-2.5, 3)$

Directrix: $x = 0.5$

Question: Given $f(x) = x^2 + 12x + 54$, find the vertex of f and the roots of f.

*Hint: it is faster to answer both questions by completing the square

Answer:

$$f(x) = x^2 + 12x + 36 + 54 - 36$$

$f(x) = (x + 6)^2 + 18$, $\therefore$ the vertex of f is $(-6, 18)$

Set $f(x) = (x + 6)^2 + 18 = 0$ and solve to find the roots of f,

The roots are $x = -6 \pm 3\sqrt{2}i$

Question: Given the general form conic equation $x^2 - 2y^2 - 6x - 4y + 25 = 0$, identify the conic section and write the equation in standard form.

Answer: Hyperbola because $A \neq C$ and they are opposite signs

$$(x^2 - 6x + 9) - 2(y^2 + 2y + 1) = -25 + 9 - 2$$

$$(x - 3)^2 - 2(y + 1)^2 = -18$$

$$\frac{(y+1)^2}{9} - \frac{(x-3)^2}{18} = 1$$

Question: Given the general form conic equation $9x^2 + 5y^2 - 20y - 160 = 0$, identify the conic section and write the equation in standard form.

Answer: Ellipse because $A \neq C$ but they are the same sign

$$9x^2 + 5(y^2 - 4y + 4) = 160 + 5(4)$$

$$9x^2 + 5(y - 2)^2 = 180$$

$$\frac{x^2}{20} + \frac{(y-2)^2}{36} = 1$$

Question: Write a matrix equation of the form $[A][X] = [B]$ that could be used to find the general form equation of a circle passing through the points (0, 4), (1, –2), and (2, 2).

Hint: Use $\begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}\begin{bmatrix} D \\ E \\ F \end{bmatrix} = \begin{bmatrix} -(x_1)^2 - (y_1)^2 \\ -(x_2)^2 - (y_2)^2 \\ -(x_3)^2 - (y_3)^2 \end{bmatrix}$

Answer: $\begin{bmatrix} 0 & 4 & 1 \\ 1 & -2 & 1 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} D \\ E \\ F \end{bmatrix} = \begin{bmatrix} -16 \\ -5 \\ -8 \end{bmatrix}$

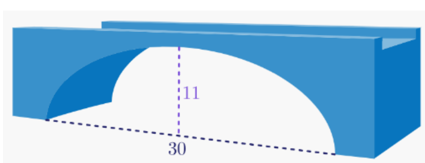
The general form equation is $5x^2 + 5y^2 + 13x - 7y - 52 = 0$

Question: An augmented matrix is given below $\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 \\ 3 & 1 & 6 & 0 & 0 & 1 \end{array}\right]$. Perform the row operations:

$R_3 = 3R_1 - R_3$: $\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 \\ 0 & -4 & -6 & 3 & 0 & -1 \end{array}\right]$

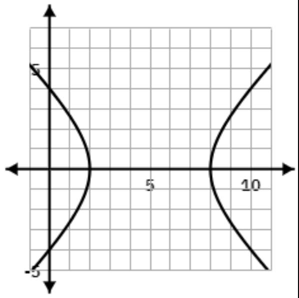
$R_2 = \frac{1}{2}R_2$: $\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 \\ 0 & -4 & -6 & 3 & 0 & -1 \end{array}\right]$

$R_3 = 4R_2 + R_3$: $\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 \\ 0 & 0 & 2 & 3 & 2 & -1 \end{array}\right]$

Graph the ellipse given by $\frac{(x+2)^2}{4}+\frac{(y-2)^2}{16}=1$ Give the length of the major axis and the coordinates of the foci	Graph the hyperbola given by $\frac{(x-5)^2}{9}-\frac{y^2}{9}=1$ Give the coordinates of the foci and the equations of the asymptotes
Describe the transformations of the quadratic parent function $y = x^2$ given by $f(x) = -(2x-8)^2 + 3$. Then graph $f(x)$ with at least three points on the curve.	Given $f(x) = -8x^2 + 8x + 3$, find a) the vertex of f b) the roots of f
A semi-elliptical archway is shown below.  Will a truck that is 8 feet wide carrying a load that reaches 6 feet above the ground safely pass through the archway? (Calculator required)	Given the general form conic equation $3x^2 - 8y + 18x + 67 = 0$ a) Identify the conic section b) Write the equation in standard form

Question: Graph the ellipse given by $\frac{(x-5)^2}{9} - \frac{y^2}{9} = 1$.
Give the coordinates of the foci and the length of the major axis.

Answer:



Asymptotes: $y = \pm(x-5)$

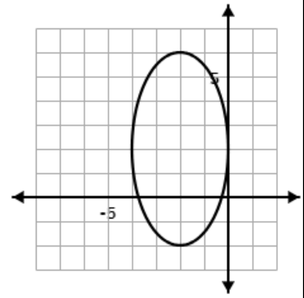
To find foci, use $c^2 = a^2 + b^2$
$= 9 + 9 = 18$

$\therefore c = 3\sqrt{2}$

So the foci are $\left(5 \pm 3\sqrt{2}, 0\right)$

Question: Graph the ellipse given by $\frac{(x+2)^2}{4} + \frac{(y-2)^2}{16} = 1$.
Give the coordinates of the foci and the length of the major axis.

Answer:



The major axis is $2a = 2(4) = 8$.

To find foci, use $c^2 = a^2 - b^2$
$= 16 - 4 = 12$

$\therefore c = 2\sqrt{3}$

So the foci are $\left(-2, 2 \pm 2\sqrt{3}\right)$

Question: Given $f(x) = -8x^2 + 8x + 3$, find the vertex of f and the roots of f.

*Good strategy is to complete the square

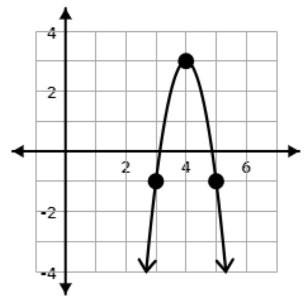
$f(x) = -8\left(x^2 - x + \frac{1}{4}\right) + 3 + 8\left(\frac{1}{4}\right)$

$f(x) = -8\left(x - \frac{1}{2}\right)^2 + 5$, $\therefore$ the vertex of f is $\left(\frac{1}{2}, 5\right)$.

Set $f(x) = -8\left(x - \frac{1}{2}\right)^2 + 5 = 0$ and solve to find the roots.

The roots are $x = \frac{1}{2} \pm \frac{\sqrt{10}}{4}$, also written as $x = \frac{2 \pm \sqrt{10}}{4}$.

Question: Describe the transformations of the quadratic parent function $y = x^2$ given by $f(x) = -(2x-8)^2 + 3$.
Then graph $f(x)$ with at least three points on the curve.



Answer:

Vertical Reflection
Horizontal Compression
Shift Right 4 units
Shift Up 3 units

Question: Given the general form conic equation $3x^2 - 8y + 18x + 67 = 0$, identify the conic section and write the equation in standard form.

Answer: Parabola because $C = 0$; there is no y^2 term.

$3(x^2 + 6x + 9) = 8y - 67 + 3(9)$

$3(x+3)^2 = 8y - 40$

$(x+3)^2 = \frac{8}{3}(y-5)$ *Note this means the focal length $p = \frac{2}{3}$

Question: A semi-elliptical archway is 30 feet wide and 11 feet tall. Will a truck that is 8 feet wide carrying a load that reaches 6 feet above the ground safely pass through the archway?

Answer: You need to find the height of the archway 4 feet away from the center; if the height is less than 6 feet, that's bad!

Equation of the semi-ellipse: $\frac{x^2}{15^2} + \frac{y^2}{11^2} = 1$

Plugging in $x = 4$ and solving for y, we find $y \sim 10.6$ feet, so yes, it is safe for the truck to pass.